Event Engine: Main Article

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- 2023-11-21 EM-61789: event traps out of order
- 2023-09-06 EM-60621: event suppression is not working
- 2023-08-28 EM-60418: Events raised against devices in a device group are not suppressed as configured
- 2022-10-24 EM-49438: Migrate /events/data_access.py to sl_events repo

- Known Issues
  - event trap long running regex expressions
  - Lack of Metrics

Introduction

Alert to Event Engine process flow: Alerts & Events Data Flow

Event Suppression and Masking: Event Suppression and Masking

Diagnosis and Mitigation

Slow Syslog Processing

This article goes into depth on this: Event Engine: Syslog event Troubleshooting

Large Inbound Emails Crashing Engine

This article goes into depth on this: Event Engine: EM7 Core: Event Processing Engine is not responding

Slow Regex Processing: find and disable policy

Event policies are somewhat like dynamic device groups: powerful features that enable customers to write badly optimized rules that can cause outages. In the case of event trap storage objects, you could

- see the full text of the regex being executed by running the storage object inspector and then
- if you see that it takes forever to run
- search for the event policy containing that regex and
- disable the policy

For example, if you knew that the regex contained the text XYZ, you could run this query:

```
select id, ename, regex_1, regex_2, regex_search from master.policies_events where regex_1 like '%XYZ%' or regex_2 like '%XYZ%;
```

From that you would have the policy ID and could then

- go through the GUI to registry → events → event manager
- set operational state to "disabled"

The regex_search column controls whether the regex_1 and regex_2 columns are interpreted as regex's (when set to 1) or just text that has to be present (when set to 0).

Event Storm

We should have a write up here for how to detect event storms and how to mitigate them.

Inspecting Event Storage Objects

We can retrieve information about event storage objects from the in_storage.hf_storage table on a CU by running the python program run_cu_query.py on the CDB. It is described in this TSG: Data Pull | inspecting storage objects with run_cu_query.py

This command line retrieves the most recent 10 storage objects that contain the string 'event' in their type name:

```
./run_cu_query.py run  "select * from in_storage.hf_storage_meta where type like '%event%' order by seq desc limit 10"
3,8394,2023-11-30 21:15:40,event_dynamic,2,event_dynamic
3,8393,2023-11-30 21:10:30,event_dynamic,2,event_dynamic
3,8392,2023-11-30 21:10:30,event_dynamic,1,event_dynamic
3,8391,2023-11-30 21:05:30,event_dynamic,2,event_dynamic
3,8390,2023-11-30 21:05:30,event_dynamic,2,event_dynamic
3,8389,2023-11-30 21:05:30,event_dynamic,2,event_dynamic
3,8388,2023-11-30 21:05:30,event_dynamic,1,event_dynamic
3,8386,2023-11-30 21:00:49,event_dynamic,2,event_dynamic
3,8385,2023-11-30 21:00:49,event_dynamic,2,event_dynamic
3,8384,2023-11-30 21:00:49,event_dynamic,2,event_dynamic

```

This is showing that, on the CU with module id 3, there are “event_dynamic” events aligned with dids 1 and 2, with sequence ids 8384, 8385, 8386, ..., 8394.

We retrieve one of these below:

```
./run_cu_query.py get_obj2 hf 3 8388  --obj-attributes=event_id,record,matching_defs
did=1, t_storage=1701378330.0
storage_db_name=None
_storage_table_name=None
did=None
data=None
gmstamp: attribute missing in object
event_id=0
record={'xtype': 1, 'seq': 4143, 'ip': '10.2.4.50', 'msg_id': 1079, 'auto_clear': 1,
'alert_id': 0, 'event_detail_uri': '', 'msg_val': '38.0', 'threshold': '15', 'message': 'CPU
runtime contention of 38.0% exceeds threshold (15%)', 'suppress_mask': datetime.timedelta(0,
600), 'Xguid': None, 'xid': 1, 'label': '', 'roa_name': '', 'log_all': 1, 'ytype': 0,
'xname': 'cdb', 'yid': 0, 'match_len': 255, 'yname': '.0', 'message_time':
datetime.datetime(2023, 11, 30, 21, 5, 23), 'active': 0, 'roa_id': 0, 'count': 1,
'msg_count': 1, 'serializationKey': 1, 'match_msg': 0}
matching_defs=[{'weight': 0, 'syslog_facility': None, 'threshold_count': 6, 'syslog_procid':
None, 'match_msg': 0, 'id': 1997, 'syslog_msgid': None, 'category': '', 'multi_match': 0,
'syslog_severity': None, 'source_host_varbind': None, 'roa_id_list': None, 'parent_id':
None, 'yname_pattern': '', 'regex_2_c': '', 'emessage': '%M', 'regex_1_t': '',
'ytype_override': None, 'xoid': 1079, 'regex_2_t': '', 'regex_search': 0, 'use_sev_mod': 0,
'regex_1_c': '', 'regex_2': '', 'trap_symbolic': None, 'regex_1': '', 'external_id': None,
'all_orgs': 1, 'yname_format': '', 'trap_oid': None, 'eseverity': 3, 'threshold_time': 30,
'syslog_appname': None}]

```

event suppression is not working

This can fail when circular relationships are involved. This happened in case EM-63893 (see section below). This TSG section explains where to find messages showing the map cycled detected by the dcmr rebuild process: DCM+R Rebuild Process | map cycles in master_dev.to po_comp_relationships .

Theory

table:cdb:master_events.events_active

```
CREATE TABLE `events_active` (
  `id` int(10) unsigned NOT NULL AUTO_INCREMENT,
  `Xtype` tinyint(3) unsigned NOT NULL DEFAULT 1,
  `Xid` bigint(20) unsigned NOT NULL DEFAULT 0,
  `Xguid` char(32) COLLATE utf8_unicode_ci DEFAULT NULL,
  `Xname` varchar(64) CHARACTER SET latin1 NOT NULL DEFAULT '',
  `Ytype` tinyint(3) unsigned NOT NULL DEFAULT 0,
  `Yid` int(10) unsigned NOT NULL DEFAULT 0,
  `Yname` varchar(512) CHARACTER SET utf8mb4 NOT NULL,
  `roa_id` mediumint(8) unsigned NOT NULL DEFAULT 0,
  `tid` mediumint(8) unsigned NOT NULL DEFAULT 0,
  `esource` tinyint(3) unsigned NOT NULL DEFAULT 1,
```

```
  `emessage` varchar(1024) CHARACTER SET utf8mb4 NOT NULL,
  `msg_val` varchar(1024) COLLATE utf8_unicode_ci DEFAULT NULL,
  `threshold` char(128) COLLATE utf8_unicode_ci DEFAULT NULL,
  `label` char(128) COLLATE utf8_unicode_ci DEFAULT NULL,
  `alert_id` int(10) unsigned DEFAULT NULL,
  `eseverity` tinyint(3) unsigned NOT NULL DEFAULT 0,
  `etype` int(10) unsigned NOT NULL DEFAULT 0,
  `ecounter` int(10) unsigned NOT NULL DEFAULT 1,
  `estate` tinyint(3) unsigned NOT NULL DEFAULT 1,
  `user_ack` int(11) unsigned NOT NULL DEFAULT 1,
  `date_ack` datetime DEFAULT '0000-00-00 00:00:00',
  `user_del` varchar(32) COLLATE utf8_unicode_ci NOT NULL,
  `date_del` datetime NOT NULL DEFAULT '0000-00-00 00:00:00',
  `date_first` datetime NOT NULL DEFAULT '0000-00-00 00:00:00',
  `date_last` datetime NOT NULL DEFAULT '0000-00-00 00:00:00',
  `date_active` datetime NOT NULL DEFAULT '0000-00-00 00:00:00',
  `notify_count` smallint(5) unsigned NOT NULL DEFAULT 0,
  `notify_stamp` timestamp NULL DEFAULT NULL,
  `suppress_time` int(10) unsigned NOT NULL DEFAULT 0,
  `suppress_group` int(10) unsigned NOT NULL DEFAULT 0,
  `suppress_topo` int(10) unsigned DEFAULT NULL,
  `force_ticket_uri` varchar(255) COLLATE utf8_unicode_ci DEFAULT NULL,
  `ext_ticket_ref` varchar(128) COLLATE utf8_unicode_ci DEFAULT NULL,
  `user_note` varchar(128) COLLATE utf8_unicode_ci NOT NULL DEFAULT '',
  `correlation_reason` varchar(256) CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL,
  `event_detail_uri` varchar(255) COLLATE utf8_unicode_ci DEFAULT NULL,
  `edit_date` datetime DEFAULT NULL,
  PRIMARY KEY (`id`),
  KEY `Xindex` (`roa_id`,`eseverity`,`etype`,`estate`,`notify_stamp`,`suppress_group`),
  KEY `estate` (`estate`),
  KEY `suppress_topo` (`suppress_topo`),
  KEY `eseverity` (`eseverity`),
  KEY `suppress_group` (`suppress_group`,`eseverity`,`id`),
  KEY `check_existing` (`Xid`,`Xtype`,`Yid`,`Ytype`,`Yname`(191),`etype`,`emessage`(128)),
  KEY `Xtype` (`Xtype`),
  KEY `date_active` (`date_active`)
) ENGINE=InnoDB AUTO_INCREMENT=17574 DEFAULT CHARSET=utf8 COLLATE=utf8_unicode_ci |
```

notes

- ID: Event ID
- Xtype: Primary element type ID
- Xid: Primary element ID

```
select * from master.definitions_xtypes;
+-------+-------------------+-------------------------------+-------------------+
| Xtype | api_resource      | Xtype_name                    | entity_class      |
+-------+-------------------+-------------------------------+-------------------+
|     0 | organization      | Organization                  | organization      |
|     1 | device            | Device                        | device            |
|     2 | asset             | Asset                         | ent_asset         |
|     4 | network           | IP Network                    |                   |
|     5 | interface         | Interface                     | ent_interface     |
|     6 | vendor            | Vendor                        | ent_vendor        |
|     7 | account           | User Account                  | account
|     8 | vinterface        | Virtual Interface             |                   |
```

```
|     9 | device_group      | Device Group                  | device_group      |
|    10 | itservice         | IT Service                    | it_service        |
|    11 | ticket            | Ticket                        | ticket            |
|    12 | cbqos_object      | Quality of Service Object     | ent_cbqos_object  |
|    13 | discovery_session | Discovery Session             | discovery_session |
|    14 |                   | Business Service              |                   |
|    15 |                   | Business Service - IT Service |                   |
|    16 |                   | Device Service                |                   |
|    17 |                   | Application                   |                   |
|    18 |                   | Application Component         |                   |
|    20 |                   | Aggregate Service             |                   |
+-------+-------------------+-------------------------------+-------------------+
19 rows in set (0.001 sec)
```

- Xguid: GUID for Primary Element (Almost always NULL)
- Xname: Name of Primary Element
- Ytype: Sub-Element type of primary element, if a subelement is used.

```
select * from master.definitions_ytypes;
+-------+-------+--------------------+--------------+
| Xtype | Ytype | Ytype_name         | api_resource |
+-------+-------+--------------------+--------------+
|     0 |     9 | News Feed          |              |
|     1 |     1 | CPU                |              |
|     1 |     2 | Disk               |              |
|     1 |     3 | File System        |              |
|     1 |     4 | Memory             |              |
|     1 |     5 | Swap               |              |
|     1 |     6 | Component          |              |
|     1 |     7 | Interface          | interface    |
|     1 |     9 | Process            |              |
|     1 |    10 | Port               |              |
|     1 |    11 | Service            |              |
|     1 |    12 | Content            |              |
|     1 |    13 | Email              |              |
|     1 |    14 | DNS                |              |
|     1 |    15 | RSS                |              |
|     1 |    16 | Quality of Service | cbqos_object |
+-------+-------+--------------------+--------------+
16 rows in set (0.001 sec)
```

- Yid and Yname: same as above, but for sub-entity
- roa_id: organization ID to which the entity belongs
- tid: ID of linked INTERNAL ticket
- esource: What triggered the event

```
select * from master.definitions_event_sources;
+---------+--------------------+
| esource | descr              |
+---------+--------------------+
|       1 | Syslog             |
|       2 | Internal           |
|       3 | Trap               |
|       4 | Dynamic            |
```

```
|        6 | API                |
|        7 | Email              |
|       10 | ScienceLogic Agent |
|       11 | Rules Engine       |
+----------+--------------------+
8 rows in set (0.001 sec)
```

- emessage: The final event message after processing. The "Event Message" in the event policy "Policy" tab. Typically %M, but can be anything.
- msg_val: The original message passed to the event policy by the esource. Could be an internal message, trap or the value passed from the DA's Alert Object's "Log Message" value depending on event source.
- threshold: The value of the %T variable defined in the DA alert formula
- label: The value of the %L variable defined in the DA alert formula
- alert_id: The alert ID that that generated the event
- eseverity: The severity of the event (backwards from what you'd expect)

```
select * from master.definitions_severities;
+----------+----------+
| severity | descr    |
+----------+----------+
|         0 | Healthy  |
|         1 | Notice   |
|         2 | Minor    |
|         3 | Major    |
|         4 | Critical |
+----------+----------+
5 rows in set (0.000 sec)
```

- etype: The event policy ID of the event that was triggered
- ecounter: The amount of times this event occurrence has triggered
- estate: The state of the event. 0: active. 1: cleared. 2: prepending.
- user_ack: The user, if any, that acknowledged the event in the event console.
- date_ack: When the event was acknowledged, if it was.
- user_del: The user that cleared the event. Can also be "auto-clear" and "expiration"
- date_del: When the event was deleted/cleared.
- date_first: The first time SL1 saw this event. Important for multiple occurrence events/pre-pending event records.
- date_last: the last time SL1 saw this event.
- date_active: When the event became active. Doesn't have to be the same time as date_first due to pre-pendings events needing multiple occurrences.
- notify_count: How many times automation has run against this event.
- notify_stamp: The last time automation was executed.

- suppress_time: something about suppression.
- suppress_group: something about suppression.
- suppress_topo: something about the type of topology that this is included in, likely related to this table:

```
select * from master.definitions_topology_types;
+------+---------+------------------------+
| type | typevar | desc                   |
+------+---------+------------------------+
|    2 | L2      | Layer 2                |
|    3 | L3      | Layer 3                |
|    5 | CDP     | CDP                    |
|    6 | LLDP    | LLDP                   |
|   10 | VM      | VMWare                 |
|  100 | ADHOC   | Ad Hoc                 |
|  101 | DCM     | Component Mapping      |
|  102 | DCMLINK | Component Relationship |
+------+---------+------------------------+
8 rows in set (0.001 sec)
```

- force_ticket_uri: External webpage that this event will link you to. Not sure where this comes into play, I forget.
- ext_ticket_ref: The linked EXTERNAL ticket number of this event. Used in different ticketing integrations to map our events to tickets in an external ticket system.
- user_note: A simple note that was attached to the ticket. Not used much. correlation_reason: used for external ticketing correlation and stuffs
- event_detail_uri: The URI to get more details about the event. Blank by default, some custom integration can use this field for tracking purposes.
- edit_date: Supposed to be the edit date of the event, but almost nothing changes this value. There are no "on update" or any other triggers that I know of that modifies this value. Unsure what it's good for. Perhaps some integration could change it

source_code:G3code/backend/events/constants.py

These constants define event states

```
ESTATE_ACTIVE = 0
ESTATE_CLEARED = 1
ESTATE_PREPENDING = 2

```

table:cdb:device_logs.logs_<did>

Device logs are generated by event policies as they execute and are written to this table:

```
CREATE TABLE `logs_0` (
  `id` int(10) unsigned NOT NULL AUTO_INCREMENT,
  `sev` tinyint(4) NOT NULL,
```

```
  `date` datetime NOT NULL,
  `message` varchar(1024) CHARACTER SET utf8mb4 NOT NULL,
  `etype` int(10) unsigned NOT NULL,
  `mtype` tinyint(4) NOT NULL,
  `e_id` int(11) NOT NULL,
  `repeats` mediumint(8) unsigned NOT NULL DEFAULT '0',
  PRIMARY KEY (`id`),
  KEY `sev` (`sev`),
  KEY `e_id` (`e_id`),
  KEY `mtype` (`mtype`),
  KEY `etype` (`etype`),
  KEY `date` (`date`)
) ENGINE=InnoDB DEFAULT CHARSET=latin1;

```

The logs for a device with id X are stored in device_logs.log_X. The click path

```
devices -> device_manager -> device wrench -> logs
```

displays a list of logs by running this query

```
SELECT
        dl.*, UNIX_TIMESTAMP(dl.date) as 'date', ea.estate
FROM
        device_logs.logs_1 dl
LEFT JOIN
        master_events.events_active ea
ON
        (dl.etype >= 1 AND dl.e_id = ea.id)
ORDER BY
        date DESC
LIMIT
        0,100;

```

and rendering the output in html. An example of this query looks like this:

```
SELECT dl.*, UNIX_TIMESTAMP(dl.date) as 'date', ea.estate FROM device_logs.logs_1 dl
LEFT JOIN master_events.events_active ea ON (dl.etype >= 1 AND dl.e_id = ea.id)  ORDER BY
date DESC LIMIT 0,100;
+------+-----+---------------------+-------------------------------------------------------
---+-------+-------+------+---------+------------+--------+
| id   | sev | date                | message
| etype | mtype | e_id | repeats | date       | estate |
+------+-----+---------------------+-------------------------------------------------------
---+-------+-------+------+---------+------------+--------+
| 1400 |   7 | 2023-11-29 20:35:20 | CPU runtime contention of 27.0% exceeds threshold (15%)
|  1997 |     4 |  326 |       0 | 1701290120 |      2 |
| 1399 |   0 | 2023-11-29 20:25:20 | CPU runtime contention of 2.0% under threshold (15%)
|  1998 |     4 |    0 |       0 | 1701289520 |   NULL |
| 1398 |   7 | 2023-11-29 20:20:27 | CPU runtime contention of 26.0% exceeds threshold (15%)
|  1997 |     4 |  326 |       0 | 1701289227 |      2 |
| 1397 |   0 | 2023-11-29 20:10:23 | CPU runtime contention of 1.0% under threshold (15%)
|  1998 |     4 |    0 |       0 | 1701288623 |   NULL |
| 1396 |   7 | 2023-11-29 20:05:20 | CPU runtime contention of 25.0% exceeds threshold (15%)
|  1997 |     4 |  326 |       0 | 1701288320 |      2 |
| 1395 |   0 | 2023-11-29 19:55:19 | CPU runtime contention of 1.0% under threshold (15%)
|  1998 |     4 |    0 |       0 | 1701287719 |   NULL |
```

```
| 1394 |   7 | 2023-11-29 19:50:25 | CPU runtime contention of 19.0% exceeds threshold (15%)
|  1997 |     4 |  326 |       0 | 1701287425 |      2 |
| 1393 |   0 | 2023-11-29 19:40:23 | CPU runtime contention of 1.0% under threshold (15%)
|  1998 |     4 |    0 |       0 | 1701286823 |   NULL |
| 1392 |   7 | 2023-11-29 19:35:21 | CPU runtime contention of 21.0% exceeds threshold (15%)
|  1997 |     4 |  326 |       0 | 1701286521 |      2 |
| 1391 |   0 | 2023-11-29 19:25:18 | CPU runtime contention of 2.0% under threshold (15%)
|  1998 |     4 |    0 |       0 | 1701285918 |   NULL |

```

This query is supposed to show messages from the events that executed the most recently at the top. However, I believe the whole second accuracy of the datetime type in this column definition

```
`date` datetime NOT NULL,
```

means that the device logs will not always appear in the order they were created. If a sequence of events executed in the event engine within the same second, and all of them inserted messages into device_logs.logs_1, they could appear out of order. I believe we could fix this by changing the order by clause in the above query to be:

```
ORDER BY
        dl.id DESC
```

It is possible to define the date column to have microsecond accuracy by changing the definition to

```
 `date` datetime(6) NOT NULL,
```

table:cdb:master_events.event_suppressions

```
CREATE TABLE event_suppressions (
  etype mediumint(9) NOT NULL DEFAULT '0',
  Xtype tinyint(3) unsigned NOT NULL DEFAULT '1',
  Xid int(11) unsigned NOT NULL DEFAULT '0',
  date_create datetime NOT NULL DEFAULT '0000-00-00 00:00:00',
  user_edit int(11) unsigned NOT NULL DEFAULT '1',
  PRIMARY KEY (etype,Xtype,Xid)
) ENGINE=InnoDB DEFAULT CHARSET=latin1;
```

db:event_insight

This database contains the following tables

```
alert_count_base.sql
alert_count_daily.sql
alert_count_hourly.sql
alert_count_raw.sql
data_aggregation_state.sql
data_rollup_state.sql
device_count_base.sql
device_count_daily.sql
device_count_hourly.sql
```

```
event_metric_base_extended.sql
event_metric_base.sql
event_metric_daily_extended.sql
event_metric_daily.sql
event_metric_hourly_extended.sql
event_metric_hourly.sql
event_metric_raw.sql
```

Some of the more important tables here, and the roles they play are: TBD

Post Mortem Articles

2024-10-25 EM-70862: Sorting is not working for the event console of the GM Stack

ts_guide_to_jira_link

Link: EM-70862: Sorting is not working for the event console of the GM Stack CLOSED

From the description:

```
Observed Behavior:
Sorting is not working for the event console of the GM Stack

Expected Behavior:
Sorting should work for the event console of the GM Stack

Summary of troubleshooting steps taken so far and the current status:
After upgrading to 12.1.2 event sorting seems to not be working properly on their GM stack. They are using the multi sorting and gql mutations mentioned on this https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1724164792372399 thread for the user they are using to check the event console as a workaround for this

 defect . They have a multi sort first for age desc and then for severity desc, sorting by age doesn't seem to work it loads older events first and then more recent ones. Before upgrading to 12.1.2 multi sorting was working,
  I ran the below gql when logged in as the user they are using to access the events console and the gql mutations are in place:

query checkUserPreferences {
userPreferences {
id
preferenceValue
}
}

I can see some unauthorized errors for a couple stacks on the event connection gql query response and in the nextui logs of the gm stack but I can't find any stack on aws with the ips mentioned,
not sure if this would affect sorting.
I can see some graphql errors in the messages logs for the EventsTableEvents  and eventCount but they just mention js scripts.
From what I can see the EventsTableEvents gql query seems to return the events and has no errors in the har file.
Discussed this issue with eng on this https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1729261571389829 slack thread.
```

2024-10-24 EM-70828: Possible issue in SL1 Interface Monitoring

ts_guide_to_jira_link

Link: EM-70828: Possible issue in SL1 Interface Monitoring CLOSED

From the description:

```
Observed Behavior: Customer not recieving clearing events for interfaces that have come back up.

Expected Behavior: Customer to get cleared or auto-cleared for interfaces that have come back up.

Summary of troubleshooting steps taken so far and the current status:
Customer will get an event when interfaces go down, but the event never clears when the interface comes back up. We captured an event and interface debug please see attatched. In the event debug logs we see
UHE "Unknown column 'rcr_severity' in 'field list'" which appears to be the same issue as https://sciencelogic.atlassian.net/browse/EM-62581#icft=EM-62581.
Interface down events with no healthy up event.
In this example once the interace is up again, we only get a successful auto-clear after the customer restarts the event engine on the collector, (This is the device example that we captured the interface and event
debug which is attatched to this escalation.)
Example device properties tab (Auto-clear Events' selected)
Example interface properties.
Major policy (No suppresions on suppressions tab)
Healthy policy (No suppresions on suppressions tab, policy is enabled)
Collector we captured the debug system_status, note the collector was built from a 12.1.0.2 ISO, it was suspected in https://sciencelogic.atlassian.net/browse/EM-62581#icft=EM-62581 that collectors built from 12.2.0 ISO only had the issue.
```

2024-08-27 EM-66572: Event Policy page slow to respond in pilot

ts_guide_to_jira_link

Link: EM-66572: Event Policy page slow to respond in pilot CLOSED

From the description:

```
Observed Behavior:Event Policy page slow to respond in New UI , but working fine in classic UI

Expected Behavior:Event policy page should work fine in new UI as well.

Summary of troubleshooting steps taken so far and the current status:
https://sciencelogic.slack.com/archives/C01GEHWGWQ3/p1724343795410329
1)We observed that Event policy page takes a long time to load with or without applying a filter in pilot. Event for cloning event policy, modifying and deletion of a event policy takes longer time .
2)We increased memory_limit to 1024 M from 128M in /etc/php.ini and restarted php-fpm services now.
3) Enabled the Debug log for nextui service and collected the log. We observed warning there
" UXXXXXP2959929W2XRXXXXXX <warn> Unknown system variable 'MAX_STATEMENT_TIME'"
4)Also collected the .HAR file log.
```

2024-07-22 EM-66021: Event Suppressions Not Working When Using Device Groups with Dynamic Rules

ts_guide_to_jira_link

Link: EM-66021: Event Suppressions Not Working When Using Device Groups with Dynamic Rules CLOSED

From the description:

```
Observed Behavior: Event suppressions do not work when using device groups with dynamic rules.

Expected Behavior: Events should be suppressed as configured.

Summary of troubleshooting steps taken so far and the current status:
Customer reports that their events are not being suppressed when the suppression is configured with a device group using dynamic rules. Even though the customer is already on 12.1.1, they applied the drop files and workaround
from the Jira https://sciencelogic.atlassian.net/browse/EM-60621 and the issue still persists. The customer is ultimately trying to get a different event policy for Device Availability for a subset of devices so the event has a
specific event message. I already worked with the customer to set this up properly but the events are still not getting suppressed properly and events for both the custom and default event policies are being triggered. I
confirmed the device we are using is part of the device group and the device is in the dynamic_rules.rule_dids_126 database table. I confirmed that when adding the device as a static device in the device group, the proper
event is suppressed and the only the desired event is triggered.

During my troubleshooting, I also saw some very strange behavior. When collecting debug logs, the event I was tracking to check the event engine debug logs would not appear, even though the event was triggering. I attempted this
several times, including once on a call, and I was unable to get the event engine debug logs to show the event I was using as an example. The event policy and example device do appear in the debug logs, but the event never does.
Requesting assistance from engineering to investigate and troubleshoot this issue further.
```

Here is the link to the previous work around: Event Engine: Main Article | 2023 09 06 EM 6062 1: event suppression is not working

2024-07-15 EM-65864: Topology suppression is not working

ts_guide_to_jira_link

Link: EM-65864: Topology suppression is not working CLOSED

From the description:

```
Observed Behavior:Topology suppression is not working

Expected Behavior:topology suppression should work and event on parent should be able to suppress child event.

Summary of troubleshooting steps taken so far and the current status:
```

```
1) The customer encountered an issue on KPN stack033 where topology suppression is not functioning correctly. Upon investigation, I found that all events had cleared, except for one event on the child device
uwvnlgor00x500. I was unable to determine the cause of this anomaly.
uwvnlgor00q001 -> parent device
uwvnlgor00x500 -> Child of uwvnlgor00q001
uwvnlgor00w005 -> Child of uwvnlgor00x500 which was suppressed via topology-suppression
uwvnlgor00w006 ->Child of uwvnlgor00x500 , and event clear is showing via auto clear
uwvnlgor00w007 ->Child of uwvnlgor00x500 , and event clear is showing via auto clear

select * from master_events.events_cleared where id in(24766394,24766396,24766377,24766375,24766391);

2) Since we lacked the necessary topology in our lab, I asked the customer to reproduce the issue in their pilot stack. Before reproducing the issue, we enabled the event engine, and observed two unusual behaviors.

A. These critical events were generated for "device failed availability" for devices within the topology. However, suppression was applied inconsistently, sometimes suppressing events on the parent device instead of the child device.

list of event ids:6391766,6391765,6391759,6391767,6391768,6391758,6391760

6391760 parent event
2411    parent device id

[root@ip-145-222-250-24 ~]# cd /var/log/sl1
[root@ip-145-222-250-24 sl1]# grep -IR "6391766"
em7_event.log:2024-07-15T13:07:43.573277+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391766 WHERE id=6391760
em7_event.log:2024-07-15T13:07:43.575045+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391766 has multiple parents: 71,71,8060
em7_event.log:2024-07-15T13:07:43.577882+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391765 WHERE id=6391766
em7_event.log:2024-07-15T13:07:43.582371+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:07:55.153899+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:08:04.107775+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:08:14.666286+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:08:24.184203+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:08:33.585967+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:08:43.684246+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
em7_event.log:2024-07-15T13:08:53.604694+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391766, this is try #2
[root@ip-145-222-250-24 sl1]# grep -IR "6391765"
em7_event.log:2024-07-15T13:07:43.573484+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391765 has multiple parents: 2411,7698
em7_event.log:2024-07-15T13:07:43.574853+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391760 WHERE id=6391765
```

```
em7_event.log:2024-07-15T13:07:43.577882+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391765 WHERE id=6391766
em7_event.log:2024-07-15T13:07:43.581028+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391765 WHERE id=6391768
em7_event.log:2024-07-15T13:07:43.582935+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391765, this is try #3
em7_event.log:2024-07-15T13:07:55.155012+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391765, this is try #3
em7_event.log:2024-07-15T13:08:04.107775+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391765, this is try #3
[root@ip-145-222-250-24 sl1]# grep -IR "6391760"
em7_event.log:2024-07-15T13:07:33.543549+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391760 WHERE id=6391758
em7_event.log:2024-07-15T13:07:33.545291+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391760 WHERE id=6391759
em7_event.log:2024-07-15T13:07:33.545555+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391760 has multiple parents: 7698
em7_event.log:2024-07-15T13:07:43.571797+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391760 has multiple parents: 7698
em7_event.log:2024-07-15T13:07:43.573277+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391766 WHERE id=6391760
em7_event.log:2024-07-15T13:07:43.574853+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391760 WHERE id=6391765
em7_event.log:2024-07-15T13:07:43.579463+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391760 WHERE id=6391767
em7_event.log:2024-07-15T13:07:43.581831+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391760, this is try #1
em7_event.log:2024-07-15T13:07:43.583446+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391760, this is try #4
em7_event.log:2024-07-15T13:07:43.583485+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 366 infinite recursion avoidance: returning 6391760
em7_event.log:2024-07-15T13:07:43.583524+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 337 Got [6391760] as the rootnode for this group
em7_event.log:2024-07-15T13:07:43.583560+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 340 Root node is already 6391760, no need to set it again
em7_event.log:2024-07-15T13:07:55.153148+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391760, this is try #1
em7_event.log:2024-07-15T13:07:55.156050+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 357 get_parent_recursive called for 6391760, this is try #4
em7_event.log:2024-07-15T13:07:55.156122+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.get_parent_recursive 366 infinite recursion avoidance: returning 6391760
em7_event.log:2024-07-15T13:07:55.156184+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 337 Got [6391760] as the rootnode for this group
e[root@ip-145-222-250-24 sl1]# grep -IR "6391758"
em7_event.log:2024-07-15T13:07:33.542052+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391758 has multiple parents: 2411,2416,7698,7698
em7_event.log:2024-07-15T13:07:33.543549+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391760 WHERE id=6391758
[root@ip-145-222-250-24 sl1]# grep -IR "6391768"
```

```
em7_event.log:2024-07-15T13:07:43.579656+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391768 has multiple parents: 8060,7698,7023,2415,2411,74
em7_event.log:2024-07-15T13:07:43.581028+00:00 localhost silo.em7_event.66[20797]: DEBUG cursor.log_query 450 QL: query:      UPDATE master_events.events_active SET suppress_topo=6391765 WHERE id=6391768
[root@ip-145-222-250-24 sl1]# grep -IR "6391767"
em7_event.log:2024-07-15T13:07:43.578090+00:00 localhost silo.em7_event.66[20797]: DEBUG shared.event_topo_suppress 319 6391767 has multiple parents: 2411,7698,7698,7698

B) When we cleared these events, the same events showing suppress_topo in the master_events.events_active table had their suppress_topo field blanked as soon as they moved to the master_events.event_cleared table.

3)i got a KB article , where https://support.sciencelogic.com/s/article/10331, Masked event shows RBA was triggered instead of the parent event and Trigger on Child Rollup is disabled
and The reason for this apparent breakdown in RBA triggering is due to the UI code using date_first to determine the displayed order of masking (who is the parent and who are the children) when severity of events is
equal*. The date_active field is more accurate because that is what determines when an RBA might be triggered, and is the true time that the event became active rather than just "possible".
But there is no solution roadmap defined for it. (attaching screenshot of this KB article as well)

4) Attaching all the logs and screenshots.
```

2024-07-02 EM-65666: [LAB] Event suppression via device group does not work

ts_guide_to_jira_link

Link: EM-65666: [LAB] Event suppression via device group does not work CLOSED

From the description:

```
Observed Behavior: The customer applied the workaround from KB article: 12176 which should be from Jira: EM-60621 which should allow for event suppressions using dynamic device groups. However, it does not appear to be working.

Expected Behavior: Suppressing a dynamic device group should work the same as using a static device group.

Summary of troubleshooting steps taken so far and the current status:
Slack discussion:
https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1719939125033579

Original description:
We recently had LAB upgraded to v12.1.1 which included known issue https://support.sciencelogic.com/s/article/12176.
The fix was applied on the 27th of June.
Testing has proven that events are still triggering RBA's even when a device group is a member of the suppression list for said event.
This needs to be fixed ASAP so that we can go ahead with our PROD upgrade tomorrow 3rd July.
```

The slack discussion is here: https://app.slack.com/client/E03R7NUFC4X/C1C2GTCAZ

The Jira case referenced is here: EM-60621: Event suppression not working CLOSED

2024-06-24 EM-65550: em7_event keeps failing and never starts after SL1 update to 12.2.0 related to trying to parse HTML tmail

ts_guide_to_jira_link

Link: 💔 EM-65550: em7_event keeps failing and never starts after SL1 update to 12.2.0 related to trying to parse HTML email CLOSED

From the description:

```
Name of Process: em7_event
Debug Logs Attached: Yes
SL1 Version Found: 12.2.0
Summary:
Event engine on DB is CRASHING and NEVER STARTING again due to unhandled exceptions with HTML parsing from an Email

The only reason this is not an active OUTAGE right now is because I found a workaround of removing the problem entries manually from SQL -- Please treat it as if it was an outage still because the issue is
going to keep recurring and needing manual intervention until it is fixed. Customer's manager is so annoyed with it they were trying to push to get SL1 reverted back to older version (which we have advised is not recommended)
Customer said this was working before upgrading to 12.2.0 and started immediately after
Getting this error when event engine tries to start:

2024-06-24T00:50:06.473593+00:00 localhost silo.em7_event.66[779822]: CRITICAL proc_pda.__init__ 260 Unhandled exception on appliance: 542, process EM7 Core:
Event Processing Engine:#012Traceback (most recent call last):#012
File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in __init__#012
File "/opt/em7/lib/python3.6/event_handler.py", line 210, in main#012
File "/opt/em7/lib/python3.6/email.py", line 373, in parse#012
File "/opt/em7/lib/python3.6/event_base.py", line 1184, in parse#012
File "/opt/em7/lib/python3.6/email.py", line 319, in _match_device#012
File "/opt/em7/lib/python3.6/email.py", line 286, in HTMLstripped#012
File "/usr/lib64/python3.6/html/parser.py", line 111, in feed#012 self.goahead(0)#012
File "/usr/lib64/python3.6/html/parser.py", line 139, in goahead#012 if self.convert_charrefs and not self.cdata_elem:#012AttributeError: 'MLStripper' object has no attribute 'convert_charrefs'

That lead me to finding this file in opengrok:
https://opengrok.sciencelogic.com/source/xref/SL1/sl_events/sl_events/email.py
I tested a slightly modified version of the failing code block in the python 3 interpreter (only modified enough to make it run in interpreter) and confirmed I am getting the same error. Even if the ONLY HTML is "<html></html>"

I also found this stackoverflow thread talking about the same issue:
https://stackoverflow.com/questions/11061058/using-htmlparser-in-python-3-2
I updated my test code with the same solution from there and it works as expected now.
Requesting a dropfile including this fix ASAP to provide to customer for testing.
Here is python interpreter output of those tests for reference:

[em7admin@clutz-dist-db01 ~]$ python3
Python 3.6.8 (default, Nov 16 2023, 04:13:44)
[GCC 8.5.0 20210514 (Red Hat 8.5.0-20.0.1)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> try:a
```

```
...     from html.parser import HTMLParser
... except:  # noqa
...     from HTMLParser import HTMLParser
...
>>> class MLStripper(HTMLParser):
...     def __init__(self):
...         self.reset()
...         self.fed = []
...     def handle_data(self, d):
...         self.fed.append(d)
...     def get_data(self):
...         return "".join(self.fed)
...
>>> def HTMLstripped(msg):
...     if "text/html" in msg:
...         print("msg is HTML-encoded: '%s'" % msg)
...     elif ("</" in msg) or ("<br>" in msg) or ("<br/>" in msg):
...         print("msg appears HTML-encoded: '%s'" % msg)
...     else:
...         print("msg not HTML-encoded, using as is: '%s'" % msg)
...         return msg  # No HTML: return as-is
...     s = MLStripper()
...     s.feed(msg.replace("<br>", " ").replace("<br/>", " "))
...     stripped_msg = s.get_data()
...     return stripped_msg
...
>>> mymsg="<html></html>"


## With ORIGINAL CLASS code

>>> HTMLstripped(mymsg)
msg appears HTML-encoded: '<html></html>'
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "<stdin>", line 10, in HTMLstripped
  File "/usr/lib64/python3.6/html/parser.py", line 111, in feed
    self.goahead(0)
  File "/usr/lib64/python3.6/html/parser.py", line 139, in goahead
    if self.convert_charrefs and not self.cdata_elem:
AttributeError: 'MLStripper' object has no attribute 'convert_charrefs'


## With the FIXED CLASS code added: "super().__init__()"

>>> class MLStripper(HTMLParser):
...     def __init__(self):
...         super().__init__()
...         self.reset()
...         self.fed = []
...     def handle_data(self, d):
...         self.fed.append(d)
...     def get_data(self):
...         return "".join(self.fed)
...
>>> HTMLstripped(mymsg)
msg appears HTML-encoded: '<html></html>'
''
```

```

Workaround:
Manually remove problem emails from the DB, like:
select seq from in_mail.messages where body like"%Notification is generated by:%";
DELETE FROM in_mail.messages where seq=1556650;

Steps to reproduce:

Get an email it doesn't like into in_mail table
```

2024-05-20 EM-65081: AP2 "Note" field in Advanced Search not filtering

ts_guide_to_jira_link

Link: EM-65081: AP2 "Note" field in Advanced Search not filtering CLOSED

From the description:

```
Reason for Escalation:
Question regarding functionality

Observed Behavior:
When using the Advanced Search in the Events page it is not possible to search Event Note.
Error Seen:
Expected "acknowledgedUser", "alignedEntityGuid", "alignedEntityId", "alignedEntityName", "alignedEntityType", "alignedResourceName", "alignedSubEntityId", "alignedSubEntityName", "alignedSubEntityType",
"counter", "dateAcknowledged", "dateActive", "dateFirst", "dateLast", "device", "eventPolicy", "extTicketRef", "id", "isAcknowledged", "message", "note", "organization", "service", "severity", "source",
"suppressGroup", "xguid", or "xtype" but "e" found.

Expected Behavior:
Based on the error it would seem like this could be expected, although I would imagine if it’s possible in the Simple Search, this should be possible in the Advanced Search
Would this be considered a Feature Request, or a Defect?
Slack Thread: https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1715687257832409
```

2024-05-20 EM-65080: AP2 event console is not working

ts_guide_to_jira_link

Link: 💔SLUI-19985: AP2 event console is not working CLOSED

From the description:

```
Hi Team,

We are seeing ScienceLogic eventing console is not working properly as we are not getting latest events on ap2 console and we are getting those alerts in classic console.
We need to fix this as soon as possible.  Will attach snapshots in next updates for reference.

Regards,
Rachna Tomar

```

```
SL1 Build: 12.1.1
Architecture: High-Availability + Disaster Recovery (HA+DR)
Remote Access: SaaS-AWS

Observed Behavior: We are seeing ScienceLogic eventing console is not working properly as we are not getting latest events on ap2 console, and we are getting those alerts in classic console.
It seems Ap2 filter is not working properly.
Expected Behavior: filter must be working properly.
Summary of troubleshooting steps taken so far and the current status:

Clean Sl1 cache.
Restarted the next Ui and nginx
Try to modify filter but no luck.
Classic UI event tab is working fine.
We try to change the Theam to default.

Note - It seems like customer having issue with Ap2 event tab filter.
```

2024-05-17 EM-65062: Require hotfix for event suppressions not working in 12.2.0 release when aligned to a device group

ts_guide_to_jira_link

Link: EM-65062: Require hotfix for event suppressions not working in 12.2.0 release when aligned to a device group CLOSED

From the description:

```
Observed Behavior:  SL1 generates events for devices that are in suppression on Dynamic device group.
Expected Behavior: SL1 does not generates events for devices that are in suppression on Dynamic device group.

Summary of troubleshooting steps taken so far and the current status:
It seems that suppressed devices in dynamic device group are not adhering to the suppression rules and are generating events randomly.

Customer version: 12.2.0, Please let me know if any logs needed to collect from DB or from collector end,

This issue is similar to the below escalation:
https://sciencelogic.atlassian.net/browse/EM-60418.

(Support Engineers need to provide a detailed summary of what steps have been taken so far and where they are currently at with the issue.)
```

2024-05-17 EM-65046: FNB Global Manager: AP2 Event Page issue Post OL8 Conversion

ts_guide_to_jira_link

Link: SLUI-19966: AP2 Event Page displays all severities mixed after loading more records CLOSED

From the description:

```
Observed Behavior: Nextui Events console page shows only critical events until you scroll down to the first Major, then it automaticly refreshes showing all severties mixed correctly.
Expected Behavior: Nextui Events console page to show all severity events mixed together.
```

2024-05-15 EM-64988: PILOT v12.1.1 after dropfile deployment- Event suppression NOT WORKING for policies with Dynamic Groups

ts_guide_to_jira_link

Link: 💔 EM-64988: PILOT v12.1.1 after dropfile deployment- Event suppression NOT WORKING for policies with Dynamic Groups on Ext Arch CLOSED

From the description:

```
Observed Behavior: When implementing the Dropfile from KB: https://support.sciencelogic.com/s/article/12176 events are still not being suppressed for All devices.
Expected Behavior: Events in Event Suppression should be suppressed.

Summary of troubleshooting steps taken so far and the current status:
KB was followed: https://support.sciencelogic.com/s/article/12176

Dropfile was implemented on Pilot (and Lab,) and didn't work. REF: https://sciencelogic.atlassian.net/browse/EM-60621#icft=EM-60621, https://sciencelogic.atlassian.net/browse/EM-63432#icft=EM-63432
Currently customer is replicating the issue and we are trying to gather the Logs from Event Engine in Debug. As soon as we have these we will upload them.
Please refer to the event policies configured for testing in attachment "EventPolicy_Group Suppression_Results.xlsx". Also the events created instead of getting suppressed "EventsCreated_Instead of Suppression_14ay2024.xlsx"
```

2024-05-15 EM-64985: Event policy Dynamic Groups suppression Dropfile v12.2.x

ts_guide_to_jira_link

Link: EM-64985: Event policy Dynamic Groups suppression Dropfile v12.2.x CLOSED

From the description:

```
Reason for Escalation:
Engineering team requested escalation
https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1715703574823859?thread_ts=1715349813.910169&cid=C1C2GTCAZ

Customer Timezone: EMEA
Versioning & Environment:
SL1 Build: 12.1.1
Architecture: High-Availability (HA)
Remote Access: No Remote Access
Salesforce Case Number: 00434633

Overview:
This is a follow on from this Esc: https://sciencelogic.atlassian.net/browse/EM-63432
```

```
Some events being generated for devices that are suppressed in Dynamic device group, particularly related to busy environments for 12.1.1, and 12.2.0/1.
In the previous escalation, a dropfile was created for v12.1.1.
KPN requires that a dropfile be validated for v12.2.0 for their planned OL8 conversion, and verified steps to install.
```

2024-04-23 EM-64573: Event Insights - Event Lifecycle shows no data

ts_guide_to_jira_link

Link: EM-64573: Event Insights - Event Lifecycle shows no data CLOSED

From the description:

```
Observed Behavior: Event Insights - Event Lifecycle is showing no data, this was working previously. Customer is unaware when it began no longer showing data.
Expected Behavior: Data to be displayed properly
Summary of troubleshooting steps taken so far and the current status:
...
Gathered debug data for processes  "Event Insight Data Aggregation" and "Event Insight Data Roll-Up"
Identified deadlocks occurring on the process in the debug
2023-09-19T02:00:15.326361+00:00 localhost silo.event_metric_aggregation.265[8771]: CRITICAL proc_pda._init_ 262 Unhandled exception on appliance: 1, process Event Insight Data Aggregation:#012Traceback (most recent call last):#012
File "/opt/em7/backend/process/proc_pda.py", line 231, in _init_#012
File "/opt/em7/backend/silo_db/event_aggregation.py", line 16, in main#012
File "/opt/em7/backend/events/data_aggregation.py", line 235, in sample#012
File "/opt/em7/backend/events/data_aggregation.py", line 164, in open_event_count_base#012
File "/opt/em7/backend/events/data_access.py", line 1426, in update_event_metrics_base#012
File "/opt/em7/backend/silo_common/database.py", line 126, in callFunc#012
File "/opt/em7/backend/silo_common/database.py", line 581, in executemany#012
File "/opt/em7/backend/silo_common/database.py", line 201, in executemany#012
File "/opt/em7/lib/python/mysqlclient-1.3.7-py2.7-linux-x86_64.egg/MySQLdb/cursors.py", line 285, in executemany#012    r = self._query(qs)#012
File "/opt/em7/lib/python/mysqlclient-1.3.7-py2.7-linux-x86_64.egg/MySQLdb/cursors.py", line 378, in _query#012    rowcount = self._do_query(q)#012
File "/opt/em7/lib/python/mysqlclient-1.3.7-py2.7-linux-x86_64.egg/MySQLdb/cursors.py", line 341, in _do_query#012    db.query(q)#012
File "/opt/em7/lib/python/mysqlclient-1.3.7-py2.7-linux-x86_64.egg/MySQLdb/connections.py", line 280, in query#012
_mysql.connection.query(self, query)#012OperationalError: (1213, 'Deadlock found when trying to get lock; try restarting transaction')
Index output of event_insight.event_metric_raw:
...
```

2024-04-10 EM-64253: Editing EventPolicy on AP2 via an AP fails

ts_guide_to_jira_link

Link: EM-64253: Editing EventPolicy on AP2 via an AP fails CLOSED

From the description:

```
Observed Behavior:Editing EventPolicy on AP2 via an AP fails

```

```
Expected Behavior:Should not through any error

Summary of troubleshooting steps taken so far and the current status:

1)Editing EventPolicy on AP2 via an AP2, gives an error message "Error while fetching event policy: Error: The "path" argument must be of type string. Received undefined"
While editing it directly via the CDB works and classic UI also works.
2)We observed few device groups were under suppression in the event policy. We have aligned each of them to "em7admin" user id.
Update master_dev.device_groups SET user_owner = <user_id> where dgid = <device_group_id>
3)We looked into  Events-->Suppressions and sort on Device/Group Name to check whether anything is appearing as "--" means historical data for already purged devices/device groups, but didnt find anything.
4)Also, while working on
,We have this behavior in 12.1 onwards that purging a device , usually not removing its entry from suppression.(master_events.events_suppressions).
5)attaching all screenshots.
```

2024-04-10 EM-64245: Multiple Unpickle errors generated during upgrade to 12.1.1

ts_guide_to_jira_link

Link: EM-64245: Multiple Unpickle errors generated during upgrade to 12.1.1 CLOSED

From the description:

```
Observed Behavior: generated many (10s of thousands) of events in the format "Storage Object Failure due to UNPICKLE ERROR"

Expected Behavior: Resolve the error

Summary of troubleshooting steps taken so far and the current status:

customer who recently upgraded to 11.3.2 to 12.1.1 and during the course of the upgrade, they generated many (10s of thousands) of events in the format "Storage Object Failure due to UNPICKLE ERROR". They
had to truncate the HF events table on all impacted collectors to allow the backlog to process.
silo.em7_hfpulld.57: Storage Object Failure due to UNPICKLE ERROR. Type: event_dynamic DID: 192215 Policy Info: event_dynamic Collector ID: 250 Row ID: 69831491 Additional Error Information: Exception
ImportError Traceback: Traceback (most recent call last): File "/opt/em7/backend/silo_db/data_pull/storage_process.py", line 202, in handle_storage_task
File "/tmp/pip-install-PbZ5ZM/sl-events/sl_events/dynamic.py", line 38, in <module>
File "/tmp/pip-install-PbZ5ZM/sl-events/sl_events/event_base.py", line 59, in <module>
File "/tmp/pip-install-PbZ5ZM/sl-events/sl_events/shared.py", line 41, in <module>
File "/tmp/pip-install-PbZ5ZM/sl-events/sl_events/data_access.py", line 9, in <module>
ImportError: cannot import name cacheme

I found similar Jira card handled by team-moss: EM-49438

```

The related issue, em49438, is linked in a section below.

2024-03-20 EM-63893 Event correlation not happening

ts_guide_to_jira_link

Link: EM-63893: Event correlation not happening CLOSED

From the description:

```
Observed Behavior: Event masking is not happening on the adhoc relationship.
Expected Behavior:
Summary of troubleshooting steps taken so far and the current status:
I have a case where the customer has reported topology masking is not working has expected. While checking their relationship we found they have implemented ad-hoc relationship as belowParent1
(Switch) - SESTL-DR07-P-SW01.amunet.com (SNMP discovered)
Child 1 (Swtich) - SESTL-DR07-p-SW02-Test.adagnet.net (SNMP discovered)
Child 2 (AP) - SESTL-DR07-P-AP01 (WLC discovered)
Child 3 (AP) - SESTL-DR07-P-AP02 (WLC discovered)But while checking child2 and child3, these are discovered WLC DCM PP with a different parent device- SESRL-SR01-V-WC02. Could you please confirm the above ad-hoc relationship
will be able to do topology suppression ?
This case has been discussed on the slack:
https://sciencelogic.slack.com/archives/C1C2GTCAZ/p1710939395300169
Please assign to Bryson Nager
```

2024-03-19 EM-63878: CDWS-SAAS-PROD: NGINX 504 Timeout errors opening Event Policies

Link: EM-63878: CDWS-SAAS-PROD: NGINX 504 Timeout errors opening Event Policies CLOSED

From the description:

```
Observed Behavior:
When accessing evnt policies through registry -> events -> event manager -> wrench an event policy, after a while it errors out with a 504 gateway timeout.
Expected Behavior:
Page should load
Summary of troubleshooting steps taken so far and the current status:
Customer suggested that we increase the php mem limit but I'm not sure that would resolve it especially since the system is already configured
to use 512Mb of PHP mem already. Plus, so far as I can tell, it's only the event policy windows that are having this issue.

[root@ip-10-152-2-100 log]# grep ^memory_limit /etc/php.d/sl1_memory_limit.ini
memory_limit = 512M

As the issue is reproducable easily, I have reproduced the issue several times to get php devlogs, or nginx logs, etc and none have shown any issues in the slightest that I can find.
I have logs and screenshots that I'll be sending to the Jira shortly.
```

2024-03-12: EM-63730: Event engine crashing frequently due to Lock wait timeout exceeded

ts_guide_to_jira_link

Link: EM-63730: Event engine crashing frequently due to Lock wait timeout exceeded CLOSED

From the description:

```
Customer has reported an issue where they are observing that event engine crashing frequently due to below error

((1205, 'Lock wait timeout exceeded; try restarting transaction')): INSERT INTO master_events.events_active (xtype, xid, xname, Xguid, ytype, yid, yname, roa_id, esource
We are not seeing high event count or HF rows behind, event engine not restarting continuously but its restarting often
We are seeing below error logs in event engine logs
==================================================
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: Traceback (most recent call last):
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/usr/lib64/python2.7/logging/handlers.py", line 863, in emit
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: self.socket.sendto(msg, self.address)
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: error: [Errno 90] Message too long
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: Logged from file proc_pda.py, line 623
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: Unhandled exception: Traceback (most recent call last):
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-process/sl_process/proc_pda.py", line 214, in _init_
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-events/sl_events/event_handler.py", line 219, in main
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-events/sl_events/events_main.py", line 167, in event_expire
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-events/sl_events/data_access.py", line 989, in remove_expired_prepending_event
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-mysql/sl_mysql/cursor.py", line 120, in wrapper
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-mysql/sl_mysql/cursor.py", line 99, in callFunc
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/tmp/pip-install-lpsIjT/sl-mysql/sl_mysql/cursor.py", line 467, in execute
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/opt/em7/lib/python/mysqlclient-1.3.7-py2.7-linux-x86_64.egg/MySQLdb/cursors.py", line 226, in execute
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: self.errorhandler(self, exc, value)
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: File "/opt/em7/lib/python/mysqlclient-1.3.7-py2.7-linux-x86_64.egg/MySQLdb/connections.py", line 36, in defaulterrorhandler
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: raise errorvalue
Feb 23 08:16:27 eu1srlpm1esdp3 event_eng.py[10658]: OperationalError: (1205, 'Lock wait timeout exceeded; try restarting transaction')
Feb 23 08:16:27 eu1srlpm1esdp3 systemd[1]: em7_event.service: main process exited, code=exited, status=1/FAILURE
Feb 23 08:16:27 eu1srlpm1esdp3 systemd[1]: Unit em7_event.service entered failed state.
=====================================================================

We have found that index_length was bit high of events_active table so we have rebuilt that but that has not made any difference, we have attached current details of event_active table
We have seen that alert_count_raw and event_metric_raw these two tables are quite a substantial amount of entries (Millions).
We are not seeing any long running queries in process list, files of locks/lockwaits are not showing anything.
Customer has failed over and rebooted the DB but issue still persist on DB.
Please help us to figure out the root cause of this issue so that event engine wont crash frequently.
```

```
We have attached all the required details and output, please let us know if anything else needs to be captured from the customer's stack.
```

2024-03-07 EM-63650: Inconsistent Clear Events API Response when using multiple filters

ts_guide_to_jira_link

Link: EM-63650: Inconsistent Clear Events API Response when using multiple filters CLOSED

From the description:

```
Observed Behavior: The API calls are getting "504 Gateway Time-Out Errors"
Expected Behavior: The API Calls should process the request.
Summary of troubleshooting steps taken so far and the current status:
(Support Engineers need to provide a detailed summary of what steps have been taken so far and where they are currently at with the issue.)
    checked the memory_limit(1024) and fastcgi_read_timeout(300)
    the customer is trying to make API call using date_active and event message for the past 12hrs
    Ex API call
    /api/cleared_event?limit=300&extended_fetch=1&filter.0.message.begins_with=Maintenance&filter.0.date_active.min=1709595059
```

2024-02-21 EM-63432: Generating events for the devices are in suppression on Dynamic device group.

ts_guide_to_jira_link

Link: EM-63432: Generating events for the devices are in suppression on Dynamic device group. CLOSED

From the description:

```
Observed Behavior: It seems that suppressed devices are not adhering to the suppression rules and are generating events randomly
Expected Behavior: Dynamic device group should suppress the all the alerts.
Summary of troubleshooting steps taken so far and the current status:
It seems that suppressed devices in dynamic device group are not adhering to the suppression rules and are generating events randomly. I found the issue on few of the servers. Attached screenshots for the reference.
Customer version: 12.1.1, Please let me know if any logs needed to collect from DB or from collector end,
This issue is similar to the below escalation:
https://sciencelogic.atlassian.net/browse/EM-60418.
```

This became a support defect, and drop files with instructions were developed and detailed in this comment: EM-63432: Generating events for the devices are in suppression on Dynamic device group. CLOSED

From this comment: EM-63432: Generating events for the devices are in suppression on Dynamic device group. CLOSED

```
Summary of findings:

 was identified as an issue by Proc and Roll team, and is under code review to be fixed for 12.2.1. This fix involves making a separate call to a different version of the get_group_dids_cached stored procedure in the case that
 this stored procedure does not return a result.

sl_events has its own version of a function call to get_group_dids_cached , and we are able to make the same change into sl_events with minimal effort.
```

2024-02-16 EM-63373: AP2 can't add device group to event policy suppression

ts_guide_to_jira_link

Link: EM-63373: AP2 can't add device group to event policy suppression CLOSED

From the description:

```
Observed Behavior: Nextui can't add device groups to event policy suppressions
Expected Behavior: To be able to in Nextui add device groups to event policy suppressions
```

2024-01-24 EM-63040: [KPN] Dynamic Device Groups added to event suppressions failing to suppress

ts_guide_to_jira_link

Link: EM-63040: [KPN] Dynamic Device Groups added to event suppressions failing to suppress CLOSED

This case was a request to backport this fix as a drop file into 11.3.1.5: EM-56787: Event is created for device if in device group for event suppression - sl_event_storage CLOSED

2023-11-21 EM-61789: event traps out of order

ts_guide_to_jira_link

Link: CMS_Vodafone SDWAN: snmp traps out of order

In this case, the device logs were out of order in the device wrench logs registry because a sequence of events that all occurred within a second of each other, created device logs. The GUI sorted these by date, which only has whole second precision. This is explained in detail here: Event Engine: Main Article | table:cdb:device_logs.logs_<did>

2023-09-06 EM-60621: event suppression is not working

ts_guide_to_jira_link

Link: EM-60621: Event suppression not working CLOSED

From the description:

```
Observed Behavior:
Events for event policies with ids 4515 and 4509 not being suppressed for devices with dids 666 and 669
even though they belong to the device group on the suppressions tab

Expected Behavior:
Events should be suppressed

Summary of troubleshooting steps taken so far and the current status:

I checked the Cisco: Power Supply Unavailable id 4515 and Cisco: Temperature Unavailable id 4509 event policies and they have the Cariad Devices and Cariad Test Device Group device groups on their suppressions.

I checked and the devices sea1-cor-swc1 did 669 and sea1-acc-swc1.net.na.cariad.vwg did 666 belong to the Cariad Devices device group but they are getting events for the Cisco: Power Supply Unavailable id 4515
and Cisco: Temperature Unavailable id 4509 event policies

Enabled debug mode for the EM7 Core: Event Processing Engine process and restarted the em7_event process on the CDB and on the collector aligned to the devices sea1-cor-swc1 did 669 and sea1-acc-swc1.net.na.cariad.vwg did 666
for debug to take effect. After this the device sea1-cor-swc1 did 669 got four Cisco: Temperature Unavailable events with ids 701247, 701240, 701237, 701234 between 22 and 23:30 EST on the 5th of September. Collected the
event logs on the CDB and on the collector, analyzed them but I couldn't find why the events are not being suppressed the CDB event logs don't contain much info and on the collector event logs I can see that the temperature
unavailable alerts are matching the events but I couldn't figure out why they aren't being suppressed.
```

2023-08-28 EM-60418: Events raised against devices in a device group are not suppressed as configured

ts_guide_to_jira_link

Link: 💔 EM-60418: Events raised against devices in a device group are not suppressed as configured CLOSED

From the description:

```
get on a webex with customer to reproduce the issue Event Suppression broke when dynamic device group is used. Need to reproduce the behavior by enabling debug on Automation and Event engine and capture debug logs. Escalate to ENGG team
Brief summary of issue and steps taken so far:
This stack has already applied the drop file for Device Group Perf
Enabled em7_automation and event engine in debugging and collected logs.
```

2022-10-24 EM-49438: Migrate /events/data_access.py to sl_events repo

ts_guide_to_jira_link

Link: EM-49438: Migrate /events/data_access.py to sl_events repo DONE

From the description:

```
Description
Migrate /events/data_access.py from em7_v8 to sl_events repository
~1596 lines
Acceptance Criteria
    data_access.py file exists within sl_events repository
    data_access.py is refactored to be compatible with both Python 2 and Python 3
    Unit tests also migrated
```

Known Issues

event trap long running regex expressions

The following post mortem details a case where data pull was brought to a halt by long a running regex, causing a HF rows behind outage: App Centrix HF Rows Behind

There are thre issues here

- the regex needs to be optimized
- policy evaluation should occur in the event engine and not be done at storage time
- we should have a way to suppress regex evaluations that go off the deep end

Long Pan indicated that, in the current framework, regex evaluation is not subject to normal event suppression but one way to introduce the ability to do that would be

- adding a timeout setting for regex evaluation

Long also looked at the possibility of moving regex evaluation out of storage and into the event engine, and it seemed plausible to him.

Lack of Metrics

Events have a creation rate and a processing rate, and when the creation rate exceeds the processing rate, the system falls further and further behind until it stops being effective at all. The event engine should be instrumented by a powerpack that gathers and plots key metrics like creation rate, processing rate, timeouts, ... so that these can be graphs can be checked during an outage.